Christine Ng

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Education

University of Washington Seattle, WA

GPA: 3.96, Bachelor of Science in Computer Science *Expected Graduation: December 2026*

Relevant Coursework: Artificial Intelligence, Deep Learning, Data Center Systems, Database System, Distributed System, Operating system, Data Structures & Algorithms

Experience

Software Development Engineer Intern June 2026 - September 2026

Amazon Web Service(AWS) - Identity and Access Management *Seattle, WA*

- Designed and deployed a policy-template validation engine using AWS Lambda and API Gateway, automating structural and semantic checks and reducing turnaround time from approximately one week to seconds.
- Integrated Amazon Bedrock to augment structured errors with contextual remediation guidance, helping users across IAM expertise levels diagnose and correct invalid policy template configurations.

Undergraduate Researcher June 2025 - Present

UW Programmable Matter Lab (Advisor: Prof. Martin Nisser) *Seattle, WA*

- Developing a computer vision and geometry-processing pipeline for a robotic wire-bending machine, converting user-uploaded images/videos of bent wire structures into editable 3D meshes for fabrication workflows.
- Developed real-time collision detection algorithm for wire–wire and wire–machine interactions, improving fabrication safety and reliability.

Leadership Experience

Lead Teaching Assistant Mar 2025 – Present

University of Washington — Data Structures and Algorithms *Seattle, WA*

- Taught weekly sections and facilitated interactive activities to reinforce core data structures and algorithmic concepts, improving student comprehension and engagement.
- Assisted in exam and assignment design, ensuring fair assessment of student understanding across the course.

Founder & Project Lead Feb 2025 - Present

VIVA - Student-Founded Event Planning Web App

- Designed and deployed backend infrastructure using Firebase (Auth, Firestore, Cloud Functions, Hosting), enabling real-time sync, secure access control, and fast RSVP updates.
- Led end-to-end technical direction for a cross-institutional engineering team, establishing coding standards, Git workflows, and UX–engineering collaboration.

Projects

Map Prefetcher | *Python, scikit-learn, JavaScript, Google Maps API, WebGazer.js*

- Developed a multimodal intent prediction system using gaze, cursor, viewport, and device-tilt signals to anticipate map navigation and prefetch tiles via real-time browser inference.
- Designed participant-held-out experiments to evaluate prediction accuracy at different time intervals before user movements, measuring prefetch latency and analyzing how individual signals complement each other using partial information decomposition.

PreVision — Multimodal Action Anticipation | *Python, PyTorch, Ego-Exo4D*

- Developing a multimodal action anticipation model using pretrained Omnivore video and BEATs audio embeddings with a causal Transformer to jointly predict the next action class and time-to-action from past observations.
- Benchmarking video-only, audio-only, and multimodal fusion models across 0.5–4-second prediction horizons to evaluate early prediction accuracy and quantify the contribution of audio features.

Technical Skills

Languages: Java, Python, C/C++, TypeScript, JavaScript, SQL (Postgres), HTML/CSS
Cloud & Backend: Docker, Kubernetes, AWS Lambda, Amazon API Gateway, Amazon Bedrock, PostgreSQL
Developer Tools: Git, Google Cloud Platform, Firebase Hosting, Blender (Python API)
Frameworks & Libraries: React, Lombok, NumPy, pandas, SciPy, scikit-learn, Matplotlib, mathutils, bmesh